

Rohith Kumar Chelluboina

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SUMMARY

Computer Science and Engineering (Artificial Intelligence & Machine Learning) graduate with hands-on experience developing AI-powered applications, intelligent agents, retrieval-augmented generation systems, and scalable backend services using Python, FastAPI, PyTorch, LangChain, and LangGraph. Experienced in designing LLM-driven workflows, AI evaluation frameworks, API development, vector search systems, and cloud-native deployments using Docker, Kubernetes, and Google Cloud Platform. Contributed to production-oriented AI systems involving multi-step reasoning, workflow orchestration, model evaluation, and intelligent automation while maintaining a strong focus on software reliability, scalability, and real-world user impact.

SKILLS

- **Programming & Web Development:** Python, Django, HTML, React, CSS, JavaScript, FastAPI, Flask, RESTful APIs
- **Databases & Cloud :** SQLite, PostgreSQL, Google Cloud Platform, FAISS, BM25 (Hybrid Search)
- **AI & Machine Learning:** AI, Machine Learning, Machine Learning Algorithms, Neural Networks, NLP, LLMs, RAG, RNNs, Pandas, NumPy, Scikit learn, PyTorch, TensorFlow, LangChain, LangGraph, Fine Tuning, Ollama, AI Agent, Prompt Engineering, Data Analysis, Data Processing
- **Tools & MLOps:** Git, Visual Studio Code, Google Colab, Jupyter Notebook, Docker, Kubernetes, CI/CD pipelines, Debugging, Model Deployment, Monitoring
- **Soft Skills:** Teamwork And Collaboration, Self Motivation, Adaptability And Flexibility, Time Management

WORK EXPERIENCE

CYPWNG Software Technologies

Mar 2025 - Sep 2025

AI Developer Intern

Hyderabad, Telangana

- Developed and maintained responsive web application features using HTML, CSS, JavaScript, Python, and Django, contributing to the Visa2Book platform's frontend and backend functionality.
- Built data preprocessing and feature engineering pipelines using Pandas and NumPy to support AI-driven recommendation and decision-support workflows.
- Integrated AI-powered recommendation systems and implemented prompt-engineering workflows using LLM technologies to enhance user guidance and automation capabilities.
- Developed and optimized FastAPI-based inference services, containerized applications with Docker, and supported scalable deployment environments.
- Created automated evaluation, testing, and regression monitoring frameworks using Python and PyTest to track model performance, reliability, and continuous improvement opportunities.
- Assisted in developing and testing AI agents using LangChain and LangGraph workflows, enabling automated reasoning, task execution, and contextual response generation across user interactions.
- Worked with structured and unstructured knowledge bases to support RAG-based AI applications, evaluating model outputs and refining prompt-engineering strategies to improve response accuracy and relevance.
- Collaborated on modular backend architecture, CI/CD workflows, and Git-based development practices to ensure maintainable, scalable, and production-ready applications.

PROJECTS

AI Memory Agent | github.com/Rohit Hubs/AI Agent.git

- Built an intelligent AI Agent using Python, FastAPI, LangChain, and transformer-based LLM workflows to automate contextual user interactions and multi-step reasoning tasks.
- Integrated Gemini APIs / AI models with backend logic to enable natural language understanding, response generation, and task execution through prompt engineering techniques.
- Implemented evaluation workflows to assess response quality, context retention, and reasoning consistency across agent interactions, supporting iterative model improvement and performance analysis.
- Integrated Gemini-based LLMs and prompt-engineering strategies to optimize conversational quality and task completion effectiveness.
- Coordinated with team members to track requirements, manage task priorities, resolve project-related queries, and ensure timely completion of deliverables within defined deadlines.

Multi-Agent Document Intelligence System | https://github.com/Rohit-Hubs/Multi_Agent_RAG_Assistant

- Architected a multi-agent RAG workflow using LangGraph to orchestrate document ingestion, context retrieval, response generation, and evaluation agents.
- Implemented Hybrid Search by combining FAISS (semantic) and BM25 (keyword) vector stores to significantly improve retrieval accuracy for complex user queries.
- Integrated Groq-powered LLMs (Llama 3.1) to generate low-latency, highly grounded responses complete with precise, page-level source citations and multi-turn conversational memory.
- Built knowledge-base ingestion and retrieval pipelines leveraging FAISS vector stores and hybrid search techniques to improve information retrieval performance.

EDUCATION

Sphoorthy Engineering College, JNTUH

B.Tech, Artificial Intelligence and Machine Learning

Sep 2023 - Apr 2026

Hyderabad, India

CERTIFICATIONS

- **Introduction to Machine Learning:** Infosys, Dec 2023
- **Introduction Artificial Intelligence:** Great Learning, Mar 2024
- **AI Developer Intern:** CYPWNG Software Technologies